But this technology is much simpler. You collect 20 nanograms of this material and explode it by giving it a thermal shock in a tiny nanomechanical switch. This thermal shock destroys the switch.

We are also working on miniaturising the micro sensor based explosive detector. We are trying to make it vapour based so that it could easily be put in cars, buses, airports, and other places. One of the ways to power these sensors could be by using the vibrations in the buses and trains. This could be done using piezoelectric methods to convert mechanical vibrations into electrical signals. We have started working on a platform of piezoelectric generators.

We are looking to use lead-free materials as piezoelectrics because lead based materials are harmful to the environment. Using applied materials, we are working on a project to develop lead-free piezoelectric materials and have filed patents for it. Now our goal is to integrate these piezoelectric harvesters into the IoT node explosive sensing. This can be discretely integrated into the existing buses without having to modify their body.

Sensors for agricultural applications

We have also started working on agricultural sensors. One of the reasons for us to get involved in this area was that the government of India has distributed a lot of soil health cards to farmers who are expected to use them to monitor the health of the soil. Farmers usually go to agricultural labs to get their soil checked. But there is no denying the fact that the whole system is faulty as there are barely any farmers who find it easy to maintain the soil health cards.

So, with the help of IIT Delhi and IIT Mumbai we have developed multiple prototypes for soil health monitoring. We are also making sensors for soil moisture measurement for which we started a company

Use of IoT in agriculture sector

called Soilsens about three years ago. We have transferred some of the technology to them and now they are commercially selling this to farmers and the others.

We are also trying to build an integrated bio-dome using artificial intelligence and machine learning for diagnostic purposes. The goal is to integrate all the non-invasive technologies that are being developed by various startups into a single bio-dome.

India is a home for great opportunities, some of which are:

- MEMS/Micro-fabrication/Batch-processing approaches ideally suited for meeting the cost and power requirements
- Address the bottom of the pyramid as most MNC products target only about 100 million of India's 1.3 billion population
- Business model required for meeting cost and maintenance requirements; calibration is a key challenge
- Local R&D for product development essential for reducing the costs and for taking care of the needs of the people in India, be it for agriculture, or security, or healthcare applications
- Multiple initiatives taken for startups by the government of India to develop technologies such as IMPRINT, UAY, GITA, BIRAC, and TSDP

The IoT technologies need to be designed in India keeping in mind the following:

The economic conditions. The techniques should be affordable for the farmers as they are not often financially sound.

Social, cultural, and educational backgrounds. Farmers are not very educated and they cannot use smartphones efficiently, unless the content is written in their local language.

Infrastructure challenges. There is a need for good infrastructure in order to calibrate these sensors in remote areas on a regular basis.

Power requirements. The farmers may not be able to change the batteries regularly, hence there is a need for a continuous power supply.

This project was founded by Intel while Nanosniffer has commercialised it. Soilsens is also selling some of these technologies today, which are multi-study projects that contribute to their efficiency. EFY

This article has been put together from a tech talk at VLSI 2022 by Prof. Ramgopal Rao of IIT Delhi. It was transcribed and curated by Laveesh Kocher, a tech enthusiast at EFY with a knack for open source exploration and research.